

Probability & Statistics Assignment 01:-

Name: Yash Raj

Roll no: 24K-0737

Q#1:-

Date:-

128	119	95	97	124	128	142	98	108	120
113	109	124	132	97	138	133	136	126	112
146	128	103	135	114	109	100	111	131	113
124	131	133	131	88	118	116	98	112	138
100	112	111	150	117	122	97	116	92	122

1) find mean

[118.62 = mean]

$$\bar{x} = \frac{\sum x_i}{n}$$

$$\bar{x} = \frac{5931}{50} = 118.62$$

2) Sort the data to find the median & mode:-

88	92	95	97	97	99	98	98	100	100
103	108	109	109	111	111	112	112	112	113
113	114	116	116	117	118	119	120	120	122
122	124	124	124	128	128	128	131	131	131
132	133	133	135	136	138	138	142	146	150

median = $n = 50$

$$\frac{50+1}{2} = \frac{51}{2} = 25.5$$

2) 25th & 26th term

$$\frac{117 + 118}{2} = 117.5$$

Ans

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c) for mode find the number with highest frequency

x	frequency
97	3
112	3
124	3
131	3

c) mode : 3

(b) find Range Variance & Standard-deviation:-

(a) Range = max-min
 $[150 - 88 = 62]$ Ans.

(b) Variance $= s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$
 $[s^2 = 165.44]$

(c) Standard-deviation $= \sqrt{s^2} = \sqrt{165.44}$
 $[s = 12.87]$

Ans

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(c) Empirical Rule:-

$$(i) \bar{y} \pm s \Rightarrow \begin{aligned} 118.62 + 12.87 &= 131.49 \\ 118.62 - 12.87 &= 105.75 \end{aligned}$$

$$(ii) \bar{y} \pm 2s \Rightarrow \begin{aligned} 118.62 + 2(12.87) &= 144.36 \\ 118.62 - 2(12.87) &= 92.88 \end{aligned}$$

$$(iii) \bar{y} \pm 3s \Rightarrow \begin{aligned} 118.62 + 3(12.87) &= 157.23 \\ 118.62 - 3(12.87) &= 80.01 \end{aligned} \quad \text{Ans.}$$

(d) Box-plot:-

$$Q_1 = \frac{n+1}{4} = \frac{51}{4} = 12.75$$

$$Q_1 = x_{12} + 0.75(x_{13} - x_{12})$$

$$Q_1 = 108 + 0.75(1)$$

$$[Q_1 = 108.75]$$

$$Q_2 = \frac{n+1}{2} = \frac{51}{2} = 25.5$$

$$= x_{25} + 0.5(x_{26} - x_{25})$$

$$= 117 + 0.5(118 - 117)$$

$$[Q_2 = 117.5]$$

$$Q_3 = \frac{3(n+1)}{4} = \frac{3(51)}{4} = 38.25$$

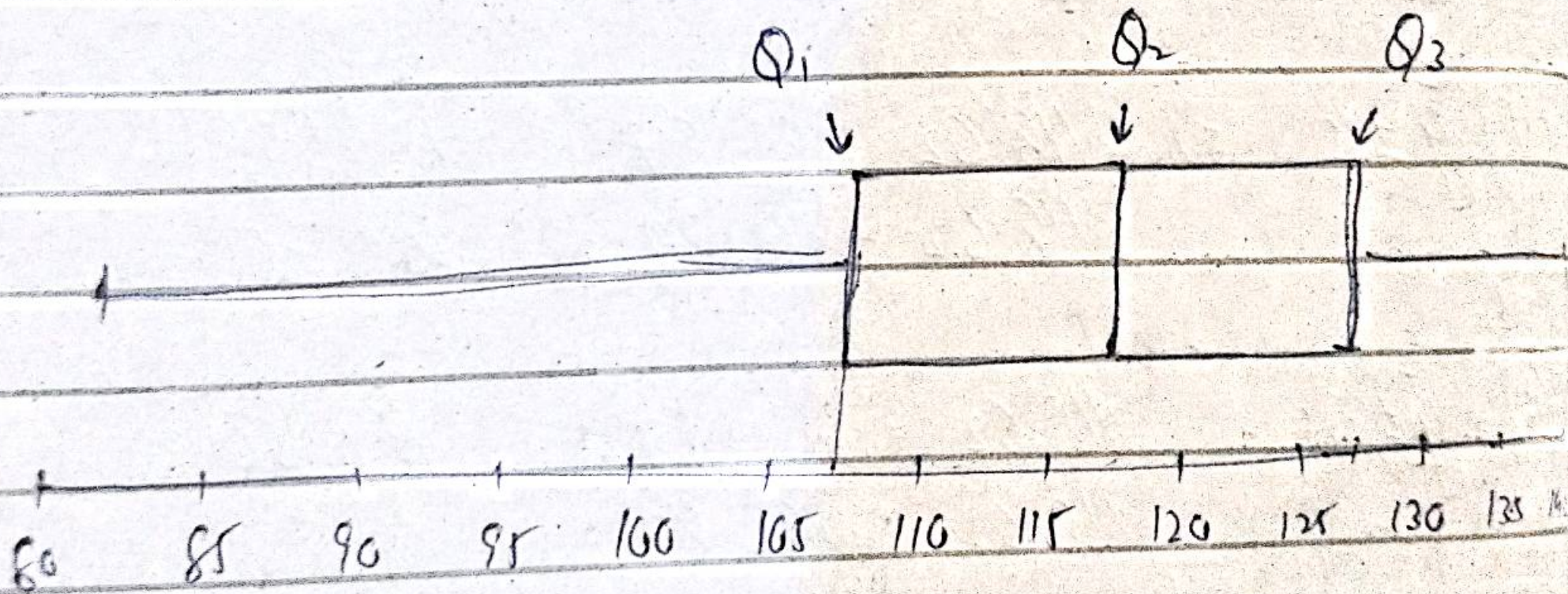
$$= x_{38} + 0.75(x_{39} - x_{38})$$

$$128 + 0.75(0)$$

$$[Q_3 = 128] \Rightarrow [Q_3 = 131]$$

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$$e) IQR, Q_3 - Q_1 \Rightarrow 131 - 108.75 \\ = 22.25$$

$$z) \text{ Upper-limit} = Q_3 + 1.5(IQR) \\ 131 + 1.5(22.25) \\ = 164.375$$

$$z) \text{ Lower-limit} = Q_1 - 1.5(IQR) \\ 108.75 - 1.5(22.25) \\ = 74.625$$

z) Since, they are in limit hence, no outliers exist.

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(e) 70th percentile:-

$$P_s = 0.7 \times (n+1)$$

$$= 0.7(51)$$

$$P_s = 36^{th}$$

$$= 124 \text{ Ans.}$$

Question #2(a): Descriptive statistics:-

$$\bar{x} = \frac{\sum f \cdot x}{\sum f} = \frac{2838}{42} = 67.57 \text{ Ans}$$

$$\text{Median: } \frac{42}{2} = 21 \quad \frac{42+1}{2} = 22$$

$$\boxed{\text{Median} = 68} \text{ Ans}$$

mode: Highest frequency:

$$\boxed{\text{mode} = 68, 77} \text{ Ans}$$

Variance & standard deviation:

$$s^2 = \frac{\sum f(x - \bar{x})^2}{\sum f}$$

$$\boxed{\text{Variance} = 260.77} \text{ Ans}$$

$$\text{Standard deviation} = \sqrt{260.77} \approx 16.15 \text{ Ans}$$

Question # 2(b):-

A: marks

B: $\mu + 6 \leq \text{marks} < \mu + 26 \Rightarrow 67.57 + 2(16.15) = 99.87$

C: $\mu \leq \text{marks} < \mu + 6 \Rightarrow 83.72 \leq \text{marks} < 99.87$

D: $\mu - 6 \leq \text{marks} < \mu \Rightarrow 67.57 \leq \text{marks} < 83.72$

E: $\mu - 26 \leq \text{marks} < \mu - 6 \Rightarrow 51.42 \leq \text{marks} < 67.57$

F: $\mu - 36 \leq \text{marks} < \mu - 26 \Rightarrow$

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A: marks $\geq \mu + 26 \Rightarrow 67.57 + 2(16.15) = 99.87$

B: $\mu + 6 \leq \text{marks} < \mu + 26 \Rightarrow 83.72 \leq \text{marks} < 99.87$

C: $\mu \leq \text{marks} < \mu + 6 \Rightarrow 67.57 \leq \text{marks} < 83.72$

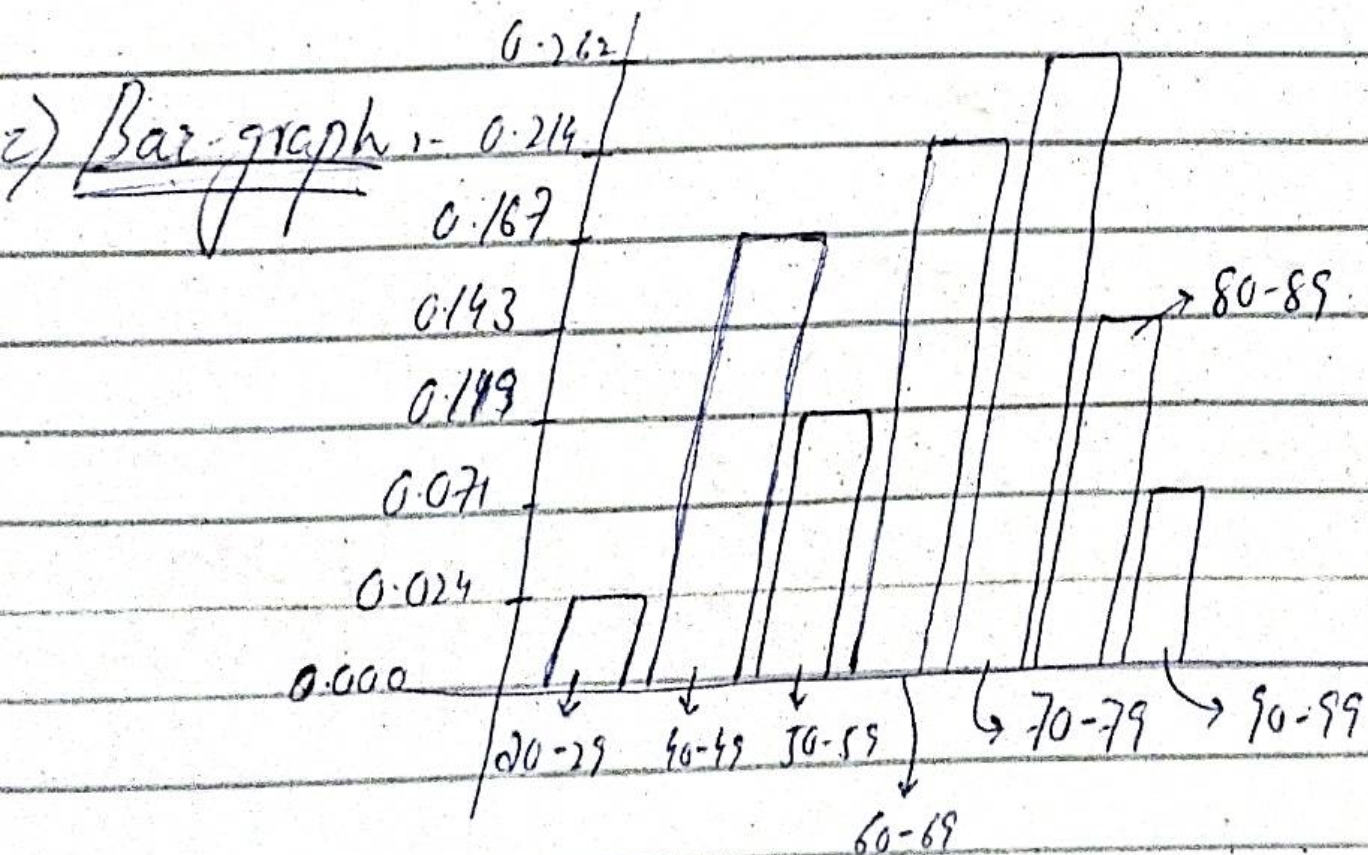
D: $\mu - 6 \leq \text{marks} < \mu \Rightarrow 51.42 \leq \text{marks} < 67.57$

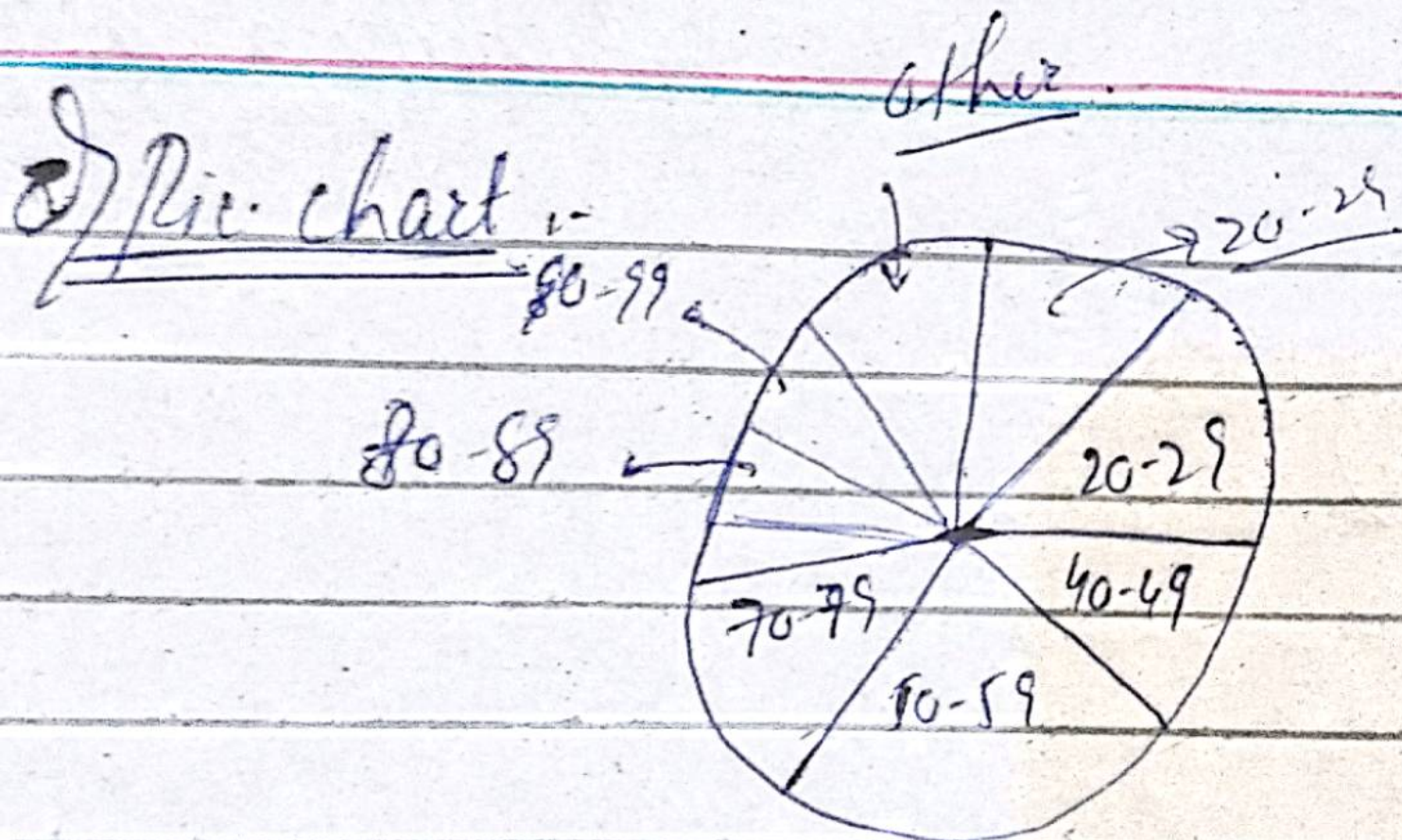
E: $\mu - 26 \leq \text{marks} < \mu - 6 \Rightarrow 35.27 \leq \text{marks} < 51.42$

F: $\mu - 36 \leq \text{marks} < \mu - 26 \Rightarrow 19.12 \leq \text{marks} < 35.27$

Question #2(c) :-

Class-Interval	frequency	relative frequency	Angle
20-29	1	0.024	8.64
30-39	0	0.000	0
40-49	7	0.167	60.12
50-59	5	0.119	42.84
60-69	9	0.214	77
70-79	11	0.262	94.32
80-89	6	0.143	51.48
90-99	3	0.071	25.56





[Signature]